

Data Diagnostics for the ENA Wildfire–CCN Study

Train/test split, distribution shift, variable structure, and model calibration

Why we care

Our causal estimates compare models trained on 2013–2024 weather against a held-out test period (six months of 2022). That comparison only means something if the test period’s weather is the *same kind* of weather the models learned on. If the test conditions have drifted, two things follow: the models may be **extrapolating** into conditions they never saw, and the causal numbers (especially ATE/ATC) become harder to trust. So before reading the causal results, this note characterizes the data: how the train/test split is built, whether the two halves share a distribution, how the input variables relate to each other and to the outcome, and whether the trained models are calibrated.

Setup. Each sample is a 241×22 back-trajectory (22 weather channels over time). For the shift and correlation tests we collapse each trajectory to 132 summary numbers (per channel: mean, spread, min, max, start, end), or to a single per-channel mean, so the tests work on interpretable vectors.

The train/test split

We use the **month-block (“paper”) split**: the test set is a fixed set of alternating months of 2022 (Jan, Mar, May, Jul, Sep, Nov); everything else across 2013–2024 is training. This mimics the reference paper and tests whether the models generalize to *held-out time periods* rather than merely interpolating between neighboring hours. Key metrics:

Metric	Train	Test
Rows (hourly samples)	57,646	3,220
Time span	2013–2024 (excl. test months)	2022 (6 alt. months)
Wildfire share (BB_criterion1)	6,378 (11.1%)	408 (12.7%)
Clean share (neither BB criterion)	51,214 (88.8%)	2,812 (87.3%)
Independent episodes (stride-24)	2,402	135

Treatment prevalence is **similar across the split** (11.1% wildfire in train vs 12.7% in test), so the split does not itself skew the wildfire/clean balance the models see.

The autocorrelation caveat. Adjacent hours share $\sim 99.7\%$ -identical back-trajectories, so the tens of thousands of rows are really only $\sim$ hundreds of *independent* episodes. Every test below is therefore run on the stride-24 decorrelated set (2,402 train / 135 test episodes), and we report effect sizes, not raw p-values, wherever autocorrelation would inflate significance.

Shift Test 1 — Classifier two-sample test (C2ST): *are they different, and how?*

Intuition. Train a classifier to guess whether a given trajectory came from the training period or the test period. If the two periods look alike, the classifier can do no better than a coin flip; if it succeeds, they differ — and *which* features it leans on tells us *what* drifted.

Result. A neural-net classifier reaches $\text{AUC} = 0.79 \pm 0.02$; a simple linear one reaches 0.73. Well above the 0.5 coin-flip line, so the periods are clearly distinguishable. The features that matter most (Figure 1) are dominated by **carbon monoxide (CO)** — a combustion tracer — followed by pressure, a temperature-variability term, and marine/aerosol-source channels.

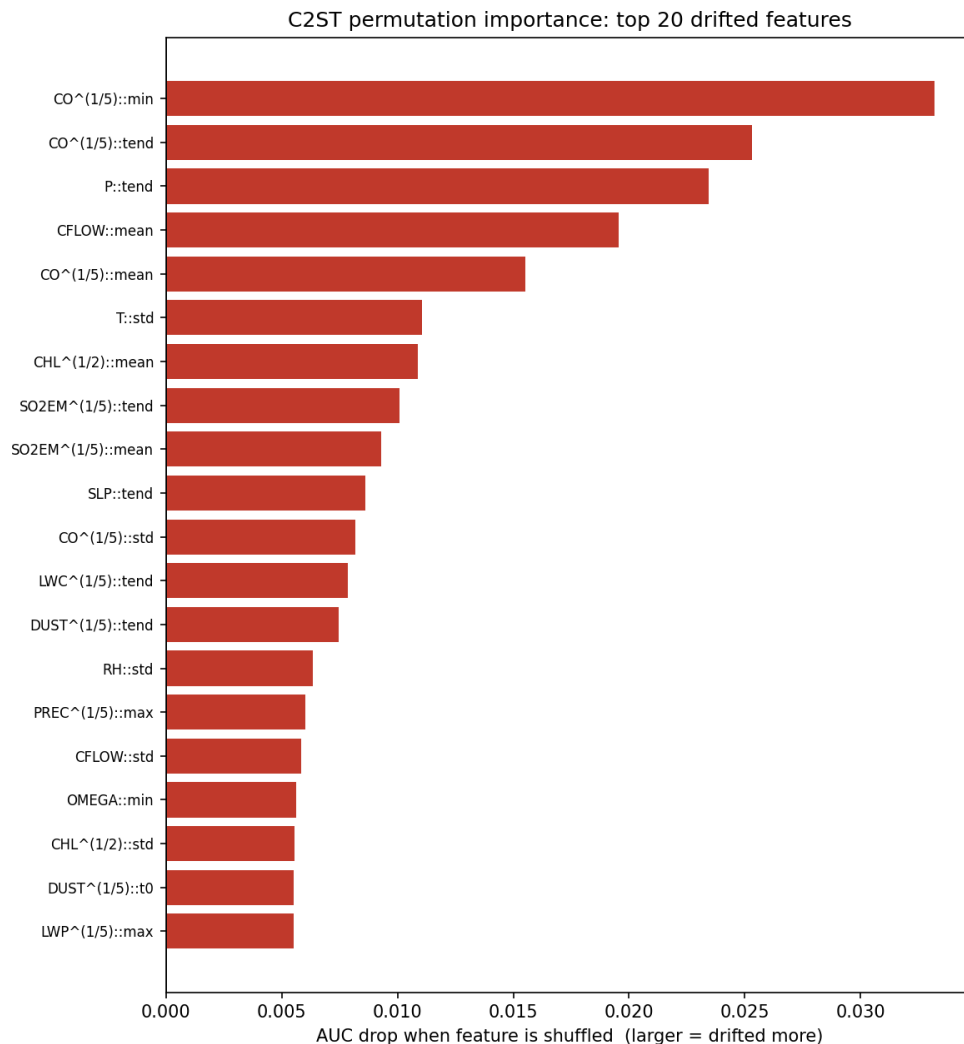


Figure 1: Which channels the classifier used to separate train from test. Longer bar = that feature drifted more. CO summaries top the list.

Shift Test 2 — MMD permutation test: *is the difference statistically real?*

Intuition. The MMD test measures how far apart the two groups are, then shuffles the “train” / “test” labels thousands of times to see how big that distance gets when there is genuinely no difference. If the real distance sits far outside that shuffled range, the difference is real.

Result. Observed $\text{MMD}^2 = 0.0072$ vs. a shuffled null centered at 0; $p = 0.0008$ (the smallest attainable with 5000 shuffles) and the observed value sits **~7 standard deviations** into the tail (Figure 2). An independent energy-distance test agrees ($p = 0.001$). Caveat: with enough independent episodes *any* nonzero difference becomes significant, so a tiny p means the shift is *real*, not *large*.

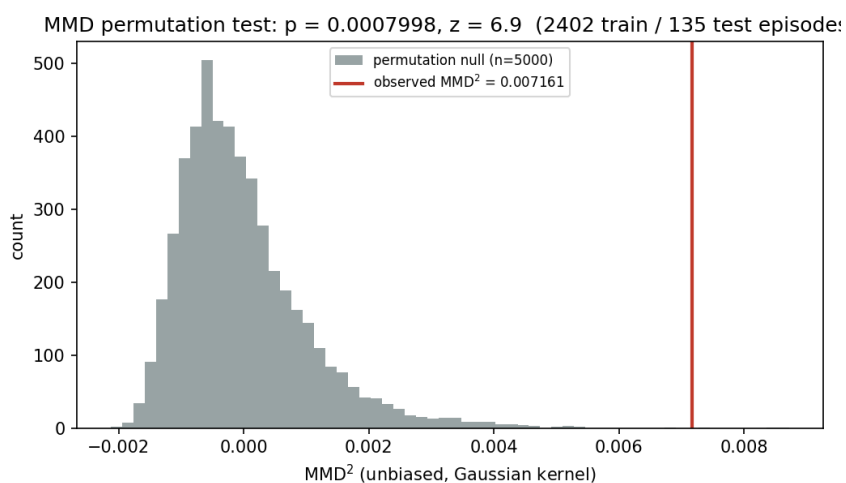


Figure 2: Grey = MMD^2 under shuffled labels (the “no difference” world). Red = the observed value; it lies far to the right of anything chance produced.

Shift Test 3 — kNN overlap: *does the test still sit inside training territory?*

Intuition. A difference in *distribution* is not the same as a move into *new* territory. For each test episode we ask: is there a training episode nearby? We compare each test point’s distance to its nearest training neighbor against how far training points sit from *each other*.

Result. Median distances are nearly equal — train→train 8.01, test→train 8.36 (**ratio 1.04**) — and only **7.4%** of test episodes fall beyond the training 95th percentile (3.7% beyond the 99th, Figure 3). Despite the significant shift, test episodes overwhelmingly land *inside* the region the models were trained on: **overlap holds**.

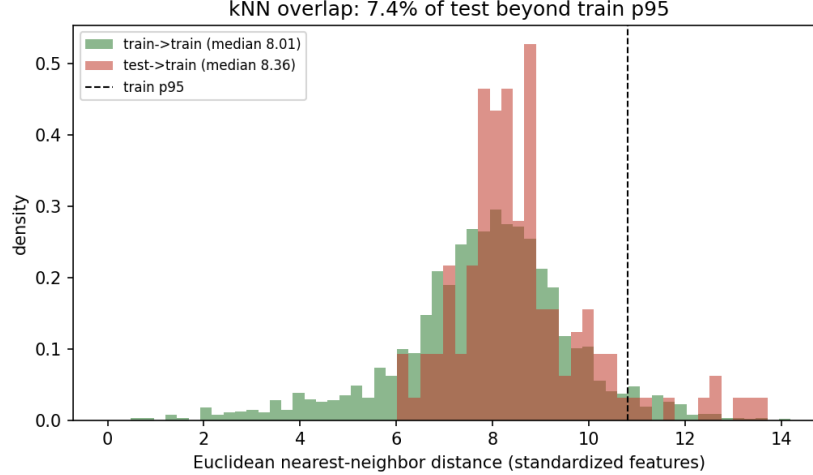


Figure 3: Distance to the nearest *training* episode. Green = training to itself; red = test to training. They overlap heavily; only a thin red tail crosses the training 95th-percentile line.

Shift verdict

Test	Question	Answer
C2ST	Are train and test different?	Yes — AUC 0.79; driven by CO
MMD	Is that difference significant?	Yes — $p = 0.0008$, $\sim 7\sigma$
kNN	Does test stay in training support?	Mostly — ratio 1.04, 7.4% past p95

The test-period weather is **genuinely and significantly different** (largely a CO / air-mass-source drift), but the difference is **modest** and the test episodes **stay within the training support**. Because overlap holds, the effect measured where both models have data (ATT) remains trustworthy, while the significant-but-real shift is why we prefer ATT over estimands (ATE/ATC) that push the models onto sparser regions.

The variables and how they relate

Beyond the split itself, it helps to see the raw structure of the 22 inputs and the outcome. Figure 4 is the full pairwise scatter matrix (every variable against every other; diagonal = each variable’s histogram), using one point per sample (the trajectory mean). Figure 5 summarizes the same relationships as correlations — *Pearson* (linear) beside *Spearman* (rank/monotonic).

The headline: **no single dominant driver of CCN**. The strongest correlate is downwelling solar radiation (SWGDN, +0.33), followed by cloud liquid-water path, low-cloud fraction, and wind speed (all ≈ -0.25). CCN rises under sunny, warm, low-cloud, calm conditions and falls with more cloud/rain/wind — a coherent “clean-marine” signature. Pearson and Spearman are nearly identical, so these *marginal* relationships are close to monotone-linear; the strong nonlinearity is in the *joint* structure, not the one-to-one marginals. Notably CO — the variable that dominates the train/test drift above — has only a weak marginal correlation with CCN, consistent with it flagging *when* wildfire air arrives rather than setting CCN on its own.

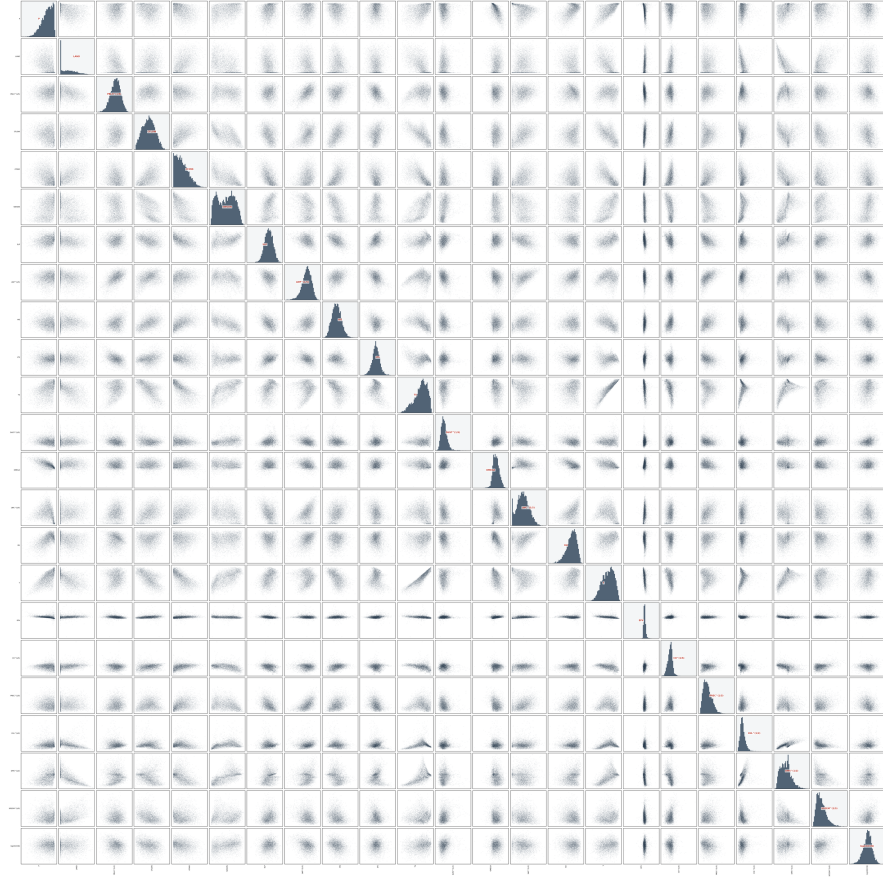


Figure 4: Pairwise scatter matrix of all 22 channels + $\log_{10}(\text{CCN})$ (bottom-right), 6000 samples, per-sample trajectory means. The bottom row / right column shows each variable against the outcome. (Downscaled here; full-resolution PDF available separately.)

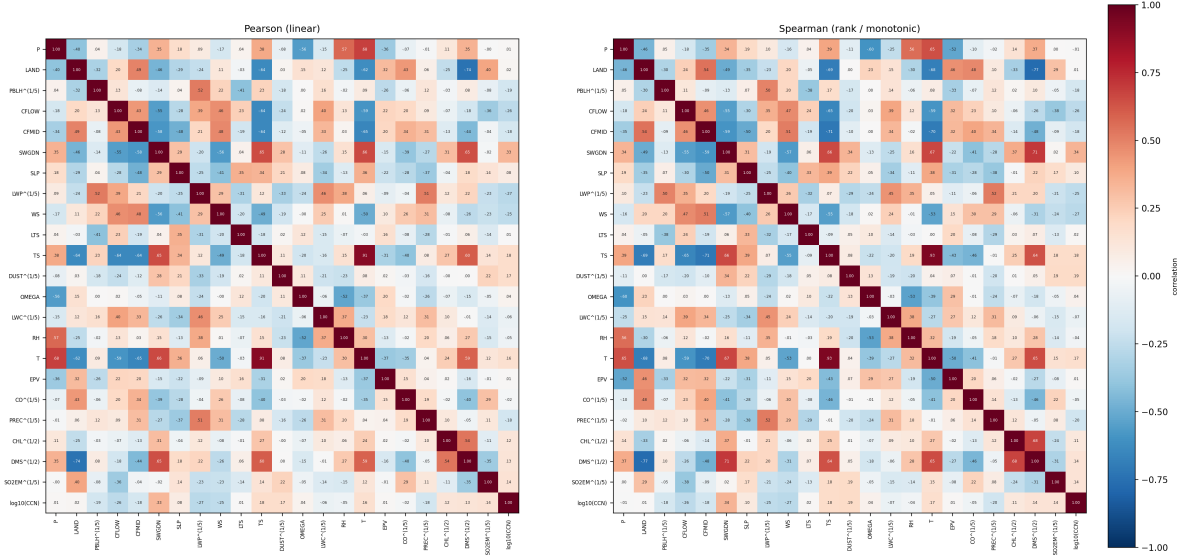


Figure 5: Correlation of all 23 variables over 20,000 samples (per-sample means). Left = Pearson (linear), right = Spearman (rank). Red = negative, blue = positive. Their near-identity indicates the marginal relationships are close to monotone-linear.

Are the models calibrated? (PIT)

Finally, a check on the stochastic generators themselves. For each held-out point we compute the **PIT** (probability integral transform): the fraction of the model’s predicted samples that fall below the observed CCN. If the predicted distribution is right, PIT values are uniform, so the histogram should be **flat**. A **U-shape** means the intervals are too narrow (overconfident); a dome means too wide. Figure 6 shows the PIT histogram for all 48 sweep configurations.

Most panels show a mild **U-shape** — the models are slightly overconfident (true CCN lands in the tails a bit more often than a calibrated model would predict) — but none are severely miscalibrated, and the better configurations are close to flat. This is the expected regime for energy-score-trained generators and is worth keeping in mind when reading interval-based quantities.

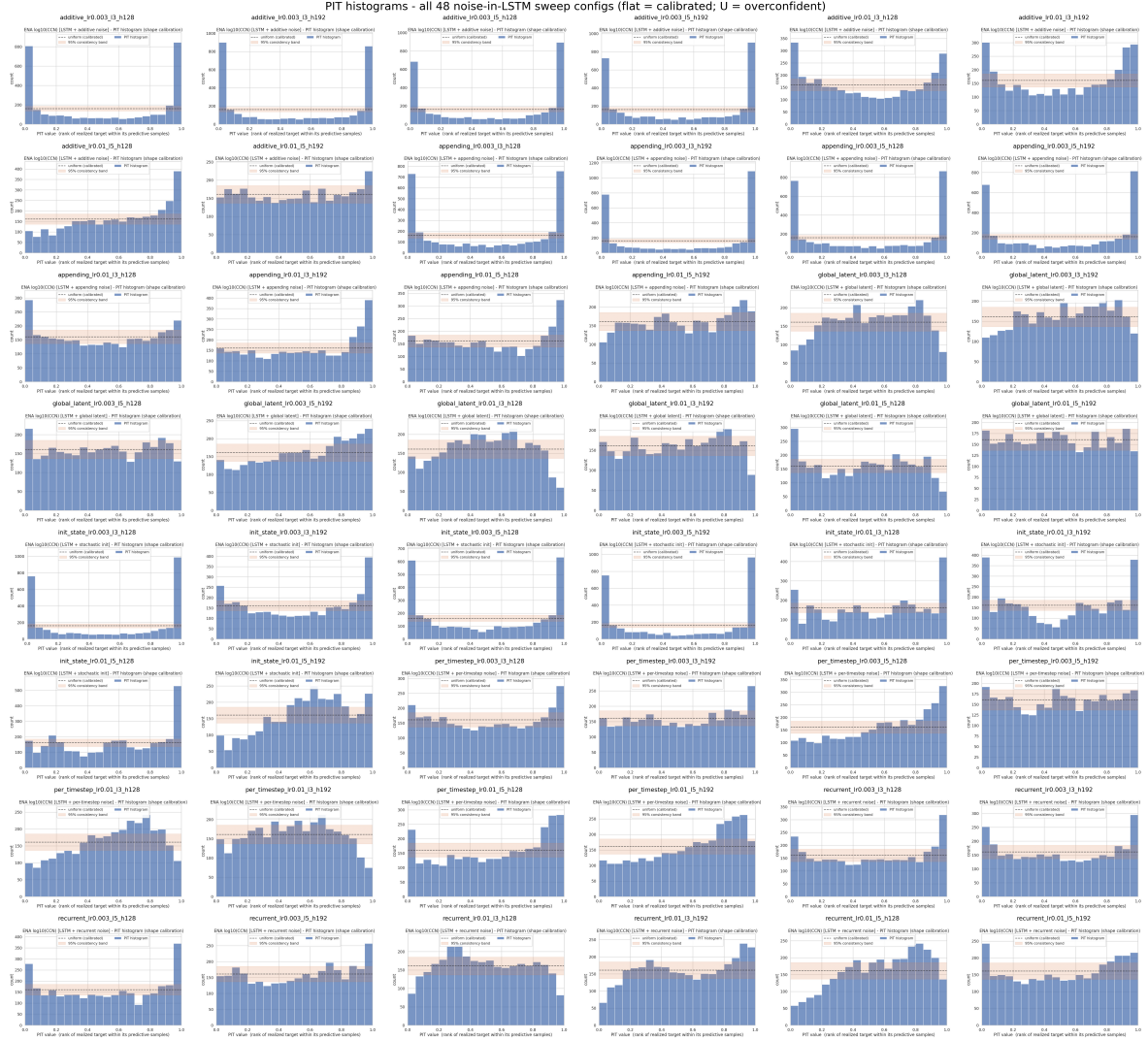


Figure 6: PIT histograms for all 48 sweep configurations (architecture $\times$ hyper-parameters), each panel titled by its config. Flat = calibrated; U-shape = overconfident (intervals too narrow); dome = underconfident.